

Vivek Venigalla

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EDUCATION

The University of Georgia, Athens, GA

- **Majors:** Mathematics with Financial Mathematics emphasis + Computer Science
- **Honors:** Classic Scholarship, Presidential Scholar Candidate, Multi-Time TSA State and National Finalist
- **Standardized Test:** ACT: 36 (Math: 36, English: 36, Reading: 35, Science: 35)
- **Relevant Coursework:** Hedging and Financial Derivatives, Microeconomics

SKILLS

- **Programming Languages:** Python, C++, Java, Haskell, R
- **Project Management:** Git, Github
- **Soft Skills:** Problem solving, prepared presentations, collaboration, adaptability

PROFESSIONAL EXPERIENCE

Eye Level, *Instructor*, Alpharetta, GA

May 2022 – November 2024

- Tutored students from grades K-8 in numerical fluency, Algebra, Pre-Calculus, English conventions and comprehension
- Organized students' personalized curriculum and timetable in folders and maintained a clean and efficient work place

LEADERSHIP & RELEVANT EXPERIENCE

Georgia Tech, *Instructional Designer*, Atlanta, GA

November 2025 – May 2026

- Created videos for the Georgia Tech Distance Math Program for Linear Algebra and Multivariable Calculus
- Employed the use of Manim, a mathematical visualization library as an add on to Python
- Communicated with current students and professors to create engaging and informative content

Mathematical Modeling

August 2025 – May 2026

- Developed mathematical models to simulate space dust trajectories in atmospheres and sports gambling consequences
- Utilized C++ to simulate Brownian motion and create Black Scholes Pricing Models
- Awarded SCUDEM Outstanding Award and M3 Honorable Mention Scholarship

Chess Club, *Founder/ Co-President*, Alpharetta, GA

August 2025 – May 2026

- Hosted weekly sessions for chess enthusiasts to play chess with one another
- Organized a local chess tournament at our high school with over 16 attendants

Midway Elementary School Robotics Club, *Mentor*, Alpharetta, GA

August 2024 – May 2026

- Provided advice to students on designing, building, coding, and documenting for the VEX IQ competition

Schoolhouse Tutoring, *Senior Tutor*, Remote

August 2025 – May 2026

- Hosted weekly sessions for chess enthusiasts to play chess with one another
- Organized a local chess tournament at our high school with over 16 attendants

PROJECTS AND AWARDS

Backtesting Engine

May 2026 – Present

- High-performance algorithmic trading backtesting engine built from scratch in C++, with a DearPyGUI frontend
- Simulates realistic brokerage rules with a optimized memory and a seamless UI that allows for endless customizability

BioBuilders - Envisioners

August 2025 – Present

- Designed two early diagnosis tests for diabetic retinopathy utilizing lateral flow assays and a AI image recognition model
- Tested in a controlled environment with professionally curated retinal images and tear buffers
- Currently publishing in the BioTreks Journal

UAV Drone

August 2022 – July 2025

- Designed, tested, and documented the process of creating a drone from scratch to complete an obstacle course
- Awarded 7th place in nationals in 2023 and 2025 during the TSA National Leadership Conference
- Awarded 1st place in states in 2024 and 3rd place in 2023 and 2025 during the TSA State Leadership Conference

Alumni Share

January 2024 – March 2024

- Full-stack Q&A platform to connect students to alumni coded in HTML/CSS/PHP
- Awarded 3rd place in state for the GASTC competition in Internet Applications

N-Objects Engine

May 2026 – Present

- Gravitational physics engine written in Haskell with a Python Plotly visualization
- Simulates various scenarios with strict energy conservation utilizing Euler Cromer and Runge-Kutta integration techniques

INTERESTS

Tennis, cricket, reading, physics